

Notes: Neuhann & Sockin (JFE, 2024)

Financial Market Concentration and Misallocation

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1 Big picture

- **Question.** How does concentration in financial markets affect real capital allocation, investment, and asset prices?
- **Core idea.** Large firms or large financial institutions are not price-takers in financial markets. When they buy or sell claims, they understand that their trades move prices.
- **Two-way feedback.** Financial price impact causes misallocation because it impairs risk sharing. Misallocation then raises price impact because it changes the cross-sectional distribution of cash flows and the residual demand faced by strategic traders.
- **Main mechanism.** High-productivity firms are natural borrowers in financial markets. Price impact makes risk sharing expensive, so they borrow less and invest less. Low-productivity firms are natural lenders; price impact makes lending less attractive, so they keep more capital in their own low-productivity projects.
- **Aggregate consequence.** The economy can feature lower average productivity, lower risky investment, more storage/cash holdings, and lower financial returns.
- **Why dispersion matters.** The distortion is stronger when productivity dispersion is high. More dispersion creates greater gains from reallocating risk and capital, so strategic trading distortions become more important.
- **Calibration exercise.** The model is calibrated to 2002 using the risk-free rate, perceived WACC, and firm discount rates. The authors then feed in the observed rise in productivity dispersion from 2002 to 2016 and a small rise in financial market concentration.
- **Quantitative message.** Rising dispersion alone can lower the risk-free rate and WACC while leaving discount rates high. Adding a small rise in concentration improves the fit for WACC and generates lower risky investment plus positive storage/cash holdings.
- **Asset-pricing implication.** Market-based costs of capital and firm-level hurdle rates need not align when large agents internalize price impact. The wedge is not simply a measurement error; it is an equilibrium valuation wedge.
- **Takeaway.** Financial market concentration can hamper allocative efficiency even in a complete-markets model. The problem is not missing securities; it is market power in trading the securities.

2 Data and measurement in the quantitative exercise

2.1 What is measured

The paper is primarily a theoretical model with a disciplined quantitative interpretation. The key mapped objects are:

- **Risk-free rate:** annualized nominal Treasury-bill rate from FRED.
- **WACC / perceived cost of capital:** survey-based perceived weighted average cost of capital from Gormsen and Huber.
- **Discount rate / hurdle rate:** self-reported corporate discount rate from Gormsen and Huber.
- **Productivity dispersion:** interquartile range of firm-level total factor productivity from Cunningham et al.
- **Market concentration:** represented in the model by the relative size of strategic agents, $\kappa = \mu/m_f$.
- **Storage:** a safe asset interpreted as cash or a low-return safe saving opportunity.

Interpretation. The model is not estimated from a large panel regression. Instead, it is calibrated to match a few central moments and then used to interpret a stylized time-series pattern: low risk-free rates, low market costs of capital, high internal hurdle rates, weak investment, and rising corporate cash.

2.2 Model objects mapped to the data

For strategic agent i , the internal discount rate is evaluated using the agent's private state prices:

$$DiscountRate_i = \frac{\sum_z \pi(z) c_{2,i}(z)}{e - c_{1,i} - \sum_z \Lambda_i(z) (y_i(z) k_i + R s_i) - k_i - s_i}. \quad (1)$$

The perceived WACC is the analogous object evaluated at market prices:

$$WACC_i = \frac{\sum_z \pi(z) c_{2,i}(z)}{e - c_{1,i} - \sum_z q(z) (y_i(z) k_i + R s_i) - k_i - s_i}. \quad (2)$$

Interpretation. The difference between $DiscountRate_i$ and $WACC_i$ is an equilibrium wedge between private valuations and market valuations. It is central to the paper because firms can rationally use high internal hurdle rates even when market-based costs of capital are low.

2.3 Productivity types in the quantitative model

The calibration uses four strategic-agent types: green stable, green volatile, red stable, and red volatile. Green and red agents have mirror-image idiosyncratic states, so their shocks are diversifiable in principle.

- Volatile firms have exposure to boom and bust shocks and a higher average productivity component.
- Stable firms have less idiosyncratic tail risk but lower productivity.
- Under perfect competition, only volatile firms invest because risk sharing eliminates the diversifiable component of their risk.
- Therefore, any investment by stable firms is a measure of misallocation induced by market power.

Interpretation. The calibration makes the model's misallocation transparent: low-productivity stable firms should not invest in the competitive benchmark, but they do when price impact deters them from lending capital to more productive volatile firms.

3 Model and empirical/theoretical design

3.1 Agents, markets, and concentration

There are two dates, $t = 1, 2$, and a finite set of states $z \in \mathcal{Z}$. There are two classes of agents:

- **Strategic agents:** large firms or institutions that internalize how their trades affect prices.
- **Competitive fringe:** atomistic price-taking agents with limited risk-bearing capacity.

Strategic agents invest in risky technologies, safe storage, and a complete set of Arrow securities. Security z pays one unit of the numeraire at date 2 in state z .

Let $a_i(z)$ be the aggregate position of strategic type i in security z , and define total strategic demand

$$A(z) = \sum_i a_i(z). \quad (3)$$

The competitive fringe position is $a_f(z)$, and market clearing is

$$A(z) + m_f a_f(z) = 0. \quad (1)$$

Interpretation. Market concentration is captured by the relative size of strategic traders, especially $\kappa = \mu/m_f$. If $\mu \rightarrow 0$, the model nests perfect competition.

3.2 Cournot–Walras equilibrium

The equilibrium concept is Cournot–Walras. Strategic agents choose quantities while taking the residual demand curve of the competitive fringe into account. They do not submit arbitrary demand schedules; instead, each large agent recognizes that changing its position changes the fringe’s consumption and hence the equilibrium state price.

Interpretation. This concept keeps the model tractable while preserving the key friction: large agents understand that their own trades move prices.

4 Key equations and propositions

4.1 Pricing functional and price impact

The competitive fringe’s first-order condition pins down the price of each Arrow security:

$$Q(A, z) = q(z) = \pi(z)u'_f(c_{2,f}(z)), \quad c_{2,f}(z) = e_{2,f}(z) - \frac{1}{m_f}A(z). \quad (2)$$

Price impact satisfies

$$\frac{\partial Q_{j,i}(A, z)}{\partial a_i(z)} = \frac{\mu}{m_f}q'(z), \quad q'(z) = \frac{\partial q(z)}{\partial A(z)} = -\pi(z)u''_f(c_{2,f}(z)) > 0. \quad (3)$$

Interpretation. If a strategic agent buys a claim on state z , the fringe must sell it, lowering the fringe’s consumption in that state. With convex marginal utility, the state’s price rises. This is the core price-impact channel.

4.2 Strategic state prices and wedges

The private state price of strategic agent i is

$$\Lambda_i(z) = \frac{\pi(z)u'(c_{2,i}(z))}{u'(c_{1,i})}. \quad (4)$$

Optimal portfolio positions satisfy

$$\Lambda_i(z) = q(z) + \frac{\mu}{m_f}q'(z)a_i(z). \quad (5)$$

Define the wedge

$$w_i(z) = \frac{\mu}{m_f}q'(z)a_i(z). \quad (6)$$

Interpretation. Buyers have positive wedges and sellers have negative wedges. Asset prices no longer align all agents’ state prices because each large trader distorts prices in the direction favorable to its own trade.

4.3 Investment and storage conditions

Risky investment and storage obey

$$\sum_z \Lambda_i(z)y_i(z) \leq 1, \quad (= \text{ if } k_i > 0), \quad (7)$$

$$\sum_z \Lambda_i(z) R \leq 1, \quad (= \text{ if } s_i > 0). \quad (8)$$

Substituting $\Lambda_i(z) = q(z) + w_i(z)$ gives

$$\sum_z q(z) y_i(z) + \sum_z w_i(z) y_i(z) = 1, \quad \sum_z q(z) R + \sum_z w_i(z) R \leq 1. \quad (9)$$

Interpretation. This is the investment wedge. Market prices alone do not summarize the private return to investment because the agent also values the investment using its own strategic-trading wedge.

4.4 Equilibrium investment result

The model implies:

- high-productivity firms invest less than under perfect competition;
- low-productivity firms invest more than under perfect competition;
- average productivity is lower;
- for sufficiently high concentration, aggregate risky investment falls;
- safe storage or cash holdings rise.

Interpretation. Market power in financial trading is not just an asset-pricing friction. It changes who invests. Productive firms are constrained by costly risk sharing, while unproductive firms retain capital because lending to productive firms moves prices against them.

4.5 Asset-pricing implications

Average Arrow prices can be written as a private state-price component minus a price-impact component:

$$q(z) = E^i[\Lambda_i(z)] - \overline{mk}^i(z). \quad (10)$$

The market-implied risk-free rate is

$$r_f^m = \left[\sum_{z \in \mathcal{Z}} q(z) \right]^{-1}. \quad (11)$$

The market risk premium can be decomposed into a classical risk-premium term and a strategic-trading premium:

$$\sum_z R_j(z) - r_f^m = -\text{Cov} \left(\frac{q(z)}{\sum_z q(z)}, R_j(z) \right) - \sum_z \frac{w_j(z)}{\sum_z q(z)} R_j(z). \quad (12)$$

Interpretation. The strategic-trading term is not a conventional covariance risk premium. It reflects the return an agent earns because its trades move prices.

5 Identification logic and quantitative design

5.1 What makes the mechanism credible

- The model is complete markets, so the results are not driven by missing securities.
- The distortions arise because large agents internalize price impact, not because they cannot trade the right claims.
- Price impact depends on the residual risk-bearing capacity of the competitive fringe.
- Misallocation feeds back into price impact by altering the distribution of output and trading needs.

Interpretation. The paper’s conceptual experiment isolates market power in financial trading as the central friction.

5.2 Why productivity dispersion is central

- Greater dispersion raises gains from risk sharing and capital reallocation.
- In competitive markets, this would increase trading without changing real allocations too much.
- In concentrated markets, greater trading needs amplify strategic incentives.
- This makes the same level of concentration more distortionary when dispersion is high.

Interpretation. Rising dispersion can make stable market concentration look more harmful over time. This is why the model can explain changing macro-finance patterns even when concentration rises only modestly.

5.3 What the calibration does and does not prove

- The calibration matches a small number of moments in 2002.
- It then evaluates the consequences of the observed 2002–2016 increase in productivity dispersion and a small increase in concentration.
- It is not a full structural estimation or a causal empirical design.
- It is a disciplined quantitative interpretation of a mechanism.

Interpretation. The evidence is best read as a plausibility exercise: financial market concentration plus rising dispersion can jointly rationalize observed trends in hurdle rates, WACC, risk-free rates, investment, and cash holdings.

6 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent original figures/tables are grouped to save space and speed comparison. All images are directly cropped from the uploaded paper.

6.1 Table 1 and Table 2: productivity process and baseline parameters

Read: Table 1:

- Green and red agents have mirror-image productivity shocks.
- Volatile firms have boom/bust exposure and a productivity gap b relative to stable firms.
- Stable firms have lower average productivity but less exposure to extreme idiosyncratic states.

Read: Table 2:

- The storage gross return is fixed at 1.
- Strategic agents have endowment $e = 3$.
- The calibrated productivity gap is $b = 0.010$.
- Distributional production risk is $\Delta = 0.163$ in the 2002 baseline.
- Relative strategic size is $\kappa = 0.388$.

Economic message. The productivity design makes misallocation observable within the model: competitive markets allocate capital to volatile high-productivity firms, while concentrated markets induce investment by stable lower-productivity firms.

Table 1

State-contingent productivity for green agents. Red agent productivity is the mirror image.

	<i>Green agent productivity</i>			
	Boom	Favorable	Unfavorable	Bust
<i>Volatile firm</i>	$Y(z_A) + b + \delta + \Delta,$	$Y(z_A) + b + \delta$	$Y(z_A) + b - \delta$	$Y(z_A) + b - \delta - \Delta$
<i>Stable firm</i>	$Y(z_A) - b + \delta,$	$Y(z_A) - b + \delta$	$Y(z_A) - b - \delta$	$Y(z_A) - b - \delta$

Table 2

Parameter choices for the baseline calibration to 2002.

Parameter	Interpretation	Value
R	Gross Return to Storage	1.000
e	Endowment of Strategic Agents	3.000
b	Productivity Gap	0.010
δ	Volatile Production Risk	0.100
Δ	Distributional Production Risk	0.163
ρ_H	Persistence of High State	0.666
ρ_L	Persistence of Low State	0.100
Y_L	Aggregate Productivity Low State	0.970
Y_H	Aggregate Productivity High State	1.243
γ, γ_f	Agent Risk Aversion	5.014
κ	Relative Size of Strategic Agents	0.388

6.2 Table 3 and Table 4: baseline fit and experiments

Read: Table 3:

- The baseline calibration exactly matches the 2002 discount rate of 15.20%, WACC of 9.65%, and risk-free rate of 1.67%.
- Model-implied aggregate investment is 5.75 and aggregate savings/storage is zero in the 2002 baseline.

Read: Table 4:

- In 2016, the data show a discount rate of 16.47%, WACC of 8.3%, and risk-free rate of 0.60%.
- Experiment 1 raises productivity dispersion from 0.163 to 0.179 and predicts WACC of 8.57% and risk-free rate of 0.63%.
- Experiment 2 also raises concentration from 0.39 to 0.41 and predicts WACC of 8.29%, risk-free rate of 0.50%, slightly lower aggregate investment, and positive storage of 0.06.

Economic message. The model can produce a falling market cost of capital without a corresponding fall in firms' internal hurdle rates. Adding mild concentration growth also generates lower real investment and higher safe storage.

Table 3

Data and model moments for the baseline calibration.

Moment	2002 Data	Model
Discount Rate	15.20%	15.20%
WACC	9.65%	9.65%
Risk-free Rate	1.67%	1.67%
Aggregate Investment	–	5.75
Aggregate Savings	–	0

Table 4

Data and model moments for the experiments. Experiment 1 considers an increase in Δ from 0.163 to 0.179. Experiment 2 additionally raises market concentration κ from 0.39 to 0.41.

Moment	2016 Data	Experiment 1	Experiment 2
Discount Rate	16.47%	15.20%	15.06%
WACC	8.3%	8.57%	8.29%
Risk-free Rate	0.60%	0.63%	0.50%
Aggregate Investment	–	5.75	5.70
Aggregate Savings	–	0	0.06

(a) Baseline 2002 moments.

(b) 2016 data and counterfactual experiments.

6.3 Figure 1 and Figure 2: comparative statics

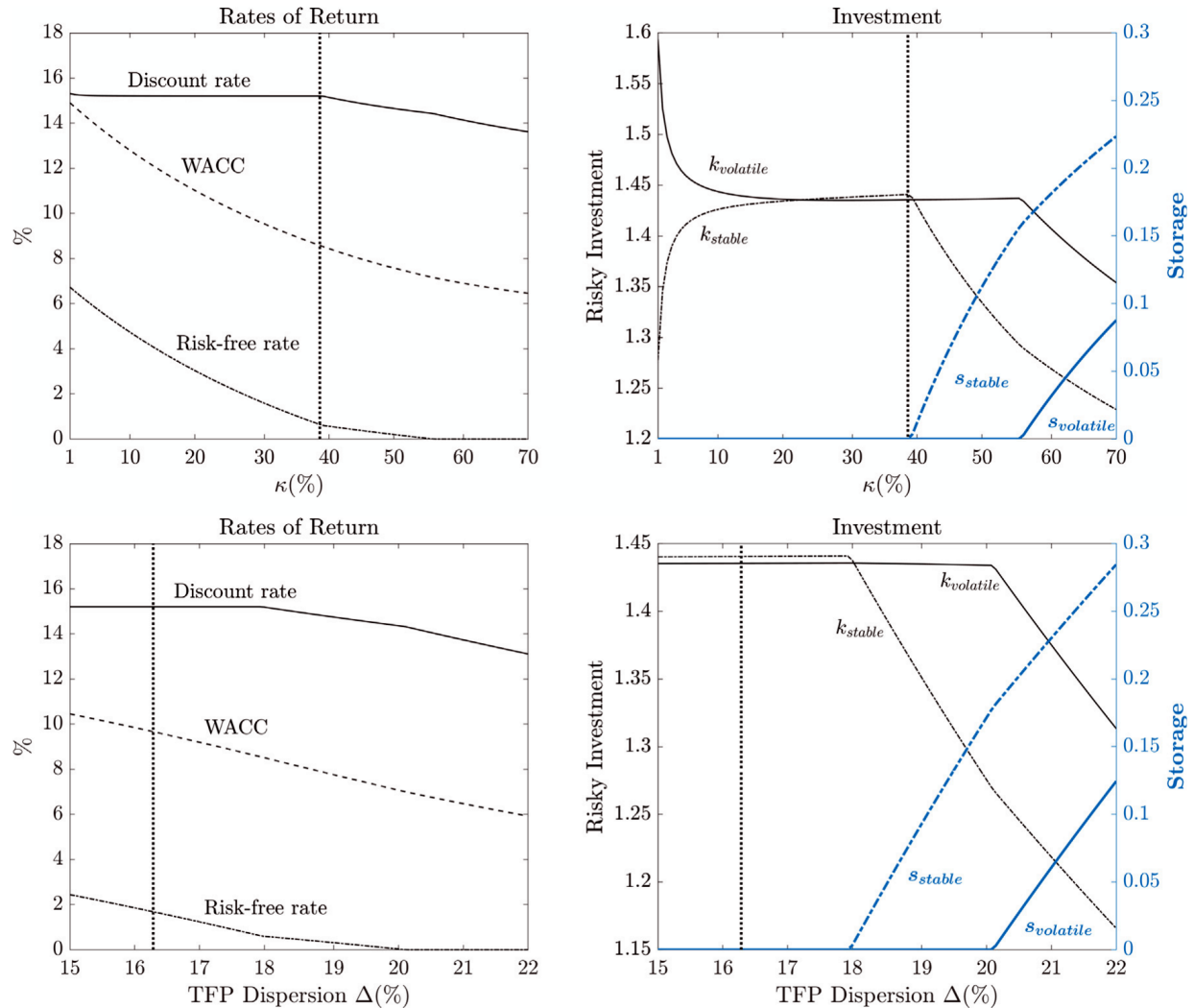
Read: Figure 1:

- Increasing concentration κ lowers the discount rate, WACC, and risk-free rate.
- High enough concentration leads firms to use storage.
- Risky investment by volatile firms eventually declines, while stable firms invest more as lending becomes less attractive.

Read: Figure 2:

- Increasing productivity dispersion Δ also lowers the risk-free rate and WACC.
- At high dispersion, firms shift toward storage and reduce risky investment.
- The comparative statics are nonlinear: initially asset-price changes dominate, but later real investment and storage responses dominate.

Economic message. The two figures illustrate the feedback loop. Concentration and dispersion both make risk sharing harder, which pushes down financial rates and eventually shifts capital away from risky productive investment.

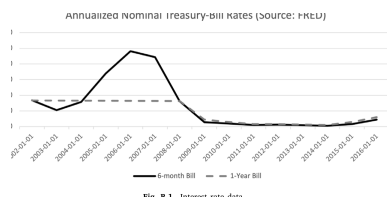


6.4 Appendix data figures: rates of return and productivity dispersion

Read:

- Figure B.1 shows the Treasury-bill rate series used to discipline the risk-free rate.
- Figure B.2 shows the discount-rate and WACC series: hurdle rates remain high while perceived WACC declines.
- Figure B.3 shows rising firm-level TFP dispersion, proxied by the interquartile range of firm-level productivity.

Economic message. The appendix figures motivate the calibration experiment. The empirical pattern is rising dispersion and a widening wedge between internal hurdle rates and market-based costs of capital.



(a) Treasury-bill rates.

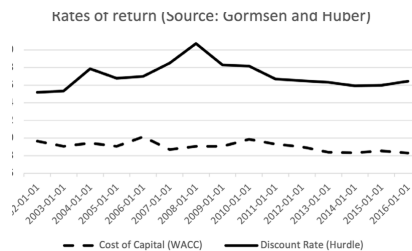


Fig. B.2. Data on cost of capital and discount rates.

(b) WACC and discount rates.

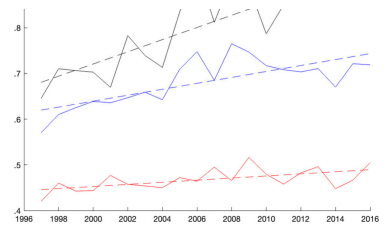


Fig. B.3. Interquartile range of firm-level TFP. Raw data are plotted in solid lines, linear trend is plotted in dashed lines.

References

(c) Firm-level TFP dispersion.

7 Short summary

- The paper studies how financial market concentration affects capital allocation.
- It builds a complete-markets model with strategic agents and a competitive fringe.
- Strategic agents internalize that their trades move prices.
- The competitive fringe pins down residual demand and state prices.
- Price impact creates wedges between private state prices and market prices.
- These wedges distort investment decisions.
- High-productivity firms under-invest because borrowing and risk sharing become expensive.
- Low-productivity firms over-invest because lending to productive firms moves prices against them.
- Misallocation lowers average productivity and can lower aggregate risky investment.
- Misallocation also raises price impact because it changes the distribution of future cash flows and trading needs.
- Rising productivity dispersion amplifies the effect of a given level of financial market concentration.
- The calibration matches 2002 discount rates, WACC, and risk-free rates.
- Feeding in the observed rise in productivity dispersion from 2002 to 2016 explains much of the decline in WACC and the risk-free rate.
- Adding a small increase in concentration produces lower risky investment and higher storage/cash holdings.
- The model explains why firms' hurdle rates may stay high even when market-based costs of capital fall.

8 Conclusion

8.1 Core conclusion

Neuhann and Sockin show that financial market concentration can create real capital misallocation even when markets are complete. Large agents internalize their price impact, distort their trades, and consequently distort their investment. This creates a feedback loop: price impact induces misallocation, and misallocation increases price impact.

8.2 Economic intuition

The central intuition is that financial markets allocate both risk and capital. When large agents have market power, the prices of risk-sharing claims move against them. High-productivity agents therefore borrow and insure less, while low-productivity agents lend less and keep capital in their own technologies. This weakens risk sharing, lowers average productivity, depresses risky investment, and can push firms toward cash-like storage.

8.3 Takeaway

The paper offers a market-power explanation for a set of post-2000 macro-finance facts: weak investment, high corporate hurdle rates, low risk-free rates, falling WACC, rising cash holdings, and growing productivity dispersion. The main message is that concentration in financial markets can be an allocative-efficiency problem, not only a liquidity or asset-pricing problem.